

Practice 3

Lesson 3 Assessment Decision Making – Build a Game Brain

Topics Covered

- Boolean expressions
- Comparison operators (`==` , `!=` , `<` , `>` , `<=` , `>=`)
- `if` statements
- `if-else` statements
- Nested `if` statements
- User input + decision making
- Basic game logic
- Score and level systems
- Simple AI/Game Brain design

Part A. Single Choice Questions (10 pts)

1. What is the purpose of an `if` statement?

- A. Repeat code many times
- B. Store data in memory
- C. Make decisions based on conditions
- D. Create functions

Answer:

2. Which operator checks whether two values are equal?

- A. `=`
- B. `==`
- C. `!=`
- D. `>=`

Answer:

3. What is the value of the following expression?

`10>5`

- A. 10

- B. 5
- C. true
- D. false

Answer:

4. What will be printed?

```
int hp =80;

if (hp>50)
{
cout<<"Alive";
}
```

- A. Nothing
- B. Alive
- C. Dead
- D. Error

Answer:

5. Which keyword executes code when the **if** condition is false?

- A. while
- B. switch
- C. else
- D. return

Answer:

6. What will be printed?

```
int score =90;
if (score>=60){
cout<<"Pass";
}else{
cout<<"Fail";
}
```

- A. Pass
- B. Fail
- C. 90
- D. Error

Answer:

7. Which of the following is a Boolean value?

- A. 42
- B. "Hello"
- C. true
- D. 3.14

Answer:

8. Which condition checks whether `age` is NOT equal to 18?

- A. `age == 18`
- B. `age = 18`
- C. `age != 18`
- D. `age >= 18`

Answer:

9. What is the output?

```
int level =3;

if (level<5)
{
    cout<<"Beginner";
}
```

- A. Beginner
- B. Advanced
- C. 3
- D. Nothing

Answer:

10. Which operator means "greater than or equal to"?

- A. `=>`
- B. `=<`
- C. `>=`
- D. `>`

Answer:

Part B. Fill in the Blanks (10 pts)

1. A condition can evaluate to either _____ or _____.

Answer:

2. Complete the condition:

```
if (age ____ 18)
```

to check if age equals 18.

Answer:

3. The keyword used when a condition is false is:

```
_____
```

Answer:

4. Complete the code:

```
if (score >= 60)
{
    cout << "Pass";
}
____
{
    cout << "Fail";
}
```

Answer:

5. What value will `isWinner` store?

```
bool isWinner = true;
```

Answer:

Part C. True or False (10 pts)

1. An `if` statement can execute different code depending on a condition.

Answer:

2. `=` and `==` mean the same thing.

Answer:

3. The expression `7 < 3` evaluates to false.

Answer:

4. A Boolean variable can store only true or false.

Answer:

5. An **else** block runs only when the **if** condition is false.

Answer:

6. The following code is valid:

```
if(score>100)
{
    cout<<"Perfect";
}
```

Answer:

7. Nested **if** statements allow decisions inside decisions.

Answer:

8. The expression **20 >= 20** evaluates to false.

Answer:

9. A game can use **if** statements to decide whether a player wins or loses.

Answer:

10. Boolean expressions are important for game AI and game logic.

Answer:

Part D. Code Reading (10 pts) Question 1

What is the output?

```
int coins =15;

if(coins>10)
{
    cout<<"Rich";
}
```

Answer

Question 2

What is the output?

```
int hp =0;

if (hp>0)
{
    cout<<"Alive";
}
else
{
    cout<<"Game Over";
}
```

Answer

Part E. Short Answer Questions (10 pts)

1. Explain the difference between `=` and `==` in C++.

Answer

2. Why are Boolean expressions important in games?

Answer

3. What is a nested `if` statement?

Sample Answer

Part F. Programming Questions (20 pts)

Question 1 – Game Health System

Ask the user to enter the player's HP.

Rules:

- $HP > 0 \rightarrow \text{"Player Alive"}$
- $HP \leq 0 \rightarrow \text{"Game Over"}$

Example

Input:

0

Output:

Game Over

Question 5 – Build a Game Brain (Challenge)

Create a mini game decision system.

Ask for:

- Player Level
- Player Coins

Rules:

```
If level >= 10 and coins >= 100:  
    VIP Shop Unlocked  
  
Otherwise:  
    Keep Grinding
```

Example

Input

```
Level: 12  
Coins: 150
```

Output

```
VIP Shop Unlocked
```

Part G. Advanced Challenge (Optional Bonus)

Rock-Paper-Scissors Judge

The computer chooses:

```
int computerChoice =1;
```

Where:

```
1 = Rock  
2 = Paper  
3 = Scissors
```

Ask the player for a choice and determine:

```
Win  
Lose
```

using `if-else` statements.

Teacher Notes

This assessment covers the core Lesson 3 concepts:

1. Boolean values (`true` / `false`)
2. Comparison operators
3. `if` statements
4. `if-else` statements
5. Nested decision making
6. Simple game logic
7. Building a "Game Brain"

The programming section gradually progresses from simple decision-making to a small game AI system, matching the theme **"Build a Game Brain"** and preparing students for future lessons on logical operators (`&&` , `||`), loops, and game state management.